Writing Technical Specifications That Get Read

A practical guide for engineering teams

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Why specifications fail

Most specifications are not wrong; they are unread. A forty-page document written to cover every case often arrives after the important decisions have already been made in chat threads and meetings. A useful specification is short enough to read in one sitting, specific enough to argue with, and written early enough to change the outcome.

1. Start with the problem

Open with two or three sentences that describe the problem in terms a new team member would understand. State who is affected, what they cannot do today, and how you will know the problem is solved. If you cannot write this section clearly, the rest of the document is premature.

2. Goals and non-goals

List what the work will achieve and, just as importantly, what it will not. Non-goals prevent scope creep and settle debates before they start. A short example for a document library:

- Goal: readers can open any PDF in the library on a phone without downloading it.
- Goal: a document's first page appears in under two seconds on a typical 4G connection.
- Non-goal: editing or annotating PDFs.
- Non-goal: offline access.

3. The proposed design

Describe the approach at the level of components and data flow, not individual functions. Include one diagram if it helps, and show the data model explicitly. Reviewers catch more problems in a table definition than in a paragraph of prose.

```
CREATE TABLE documents (
  id          TEXT PRIMARY KEY,
  title       TEXT NOT NULL,
  file_url    TEXT NOT NULL,
  page_count  INTEGER,
  created_at  TEXT NOT NULL
);
```

4. Alternatives considered

For each serious alternative, write a short paragraph on why it was rejected. This is the section future readers value most, because it answers the question every newcomer eventually asks: why did we not just do it the obvious way?

5. Risks and open questions

Be honest about what you do not know. An open question written down invites the right person to answer it; an open question left out invites a surprise in production.

- What happens to existing links when a document is renamed?
- Who can publish, and how is that access granted?
- What is the largest file we need to support?

6. Rollout and rollback

Explain how the change reaches users: feature flags, migration steps, and how to undo them. A plan that cannot be reversed deserves extra review before anyone starts building it.

A checklist before sharing

- The problem statement fits in three sentences.
- Every goal is measurable.
- Non-goals are listed.
- At least one alternative is described and rejected with reasons.
- Each open question names who should answer it.
- The document can be read in under fifteen minutes.

A specification is a tool for thinking together. Its success is measured by the quality of the conversation it starts, not by its length.